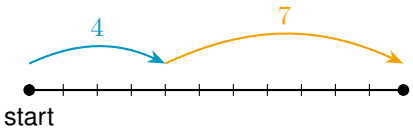
Segment Addition Postulate — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: Walk 4 steps, then 7 more*

Key Idea

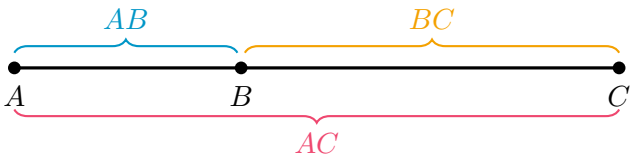


Walk 4, then 7 more.

Distance from the start: $4 + 7 =$ 11

☞ *Video: The postulate*

Key Idea



Segment Addition Postulate: If B is **between** A and C, then $AB + BC = AC$.

In words: part + part = **whole**.

☞ *Video: Simple cases first*

Key Idea

Find the whole: add

$AB = 4,\ BC = 7$

$AC = AB + BC = 4 + 7 =$ 11

Know the whole and one piece: subtract

$AC = 20,\ AB = 8$

$BC = AC - AB = 20 - 8 =$ 12

☞ *Video: With algebra*

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Watch Me — Worked Example

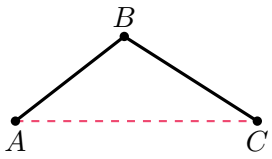
$AB = x + 3, \quad BC = 2x, \quad AC = 21.$ Find x and BC.

$(x + 3) + 2x = 21$	part + part = **whole**
$__3x + 3__ = 21$	combine like terms
$3x = __18__$	subtract 3
$x = __6__$	**don't stop at x!**
$BC = 2(6) = __12__$	answer the question they asked

Check: $AB = __9__, \ 9 + 12 = 21$ ✓

☞ *Video: Only when B is between A and C*

Common Mistake



The postulate only works when B is **between** A and C on one line.

If A, B, C make a triangle, then $AB + BC$ is **more** than AC.

☞ *Video: Your turn*

Your Turn

Q is between P and R. $\quad PQ = 3x - 1, \quad QR = x + 5, \quad PR = 24.$ Find x and PQ.

$(3x - 1) + (x + 5) = 24$

$__4x + 4__ = 24 \quad \Rightarrow \quad x = __5__$

$PQ = 3(5) - 1 = __14__$ Check: $QR = __10__, \ 14 + 10 = 24$ ✓

Notice

Remember: **part** + **part** = **whole** ($AB + BC = AC$ when B is between A and C). Missing piece? **Subtract** from the whole. Solve for x — then answer the question they actually asked.

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